

Applications of abacus diagrams: Simultaneous core partitions, alcoves, and a major statistic

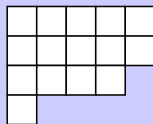
Christopher R. H. Hanusa
Queens College, CUNY

Joint work with Brant Jones, James Madison University
and Drew Armstrong, University of Miami

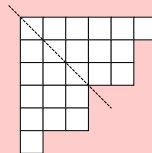
people.qc.cuny.edu/chanusa > Talks

Partitions

The **Young diagram** of $\lambda = (\lambda_1, \dots, \lambda_k)$ has λ_i boxes in row i .



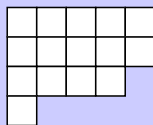
Partition



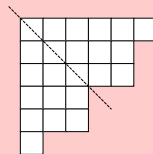
Self-conjugate partition

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 Abacus: Function $a : \mathbb{Z} \rightarrow \{\bullet, _ \}$.



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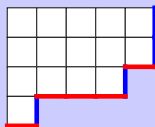
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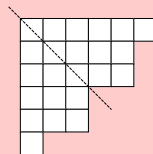
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Partitions correspond to abacus diagrams.

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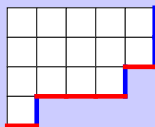
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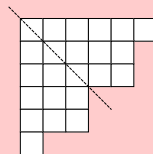
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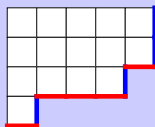
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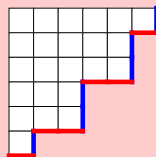
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Partition



Self-conjugate partition

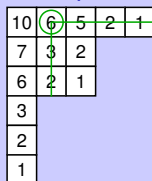
Self-conjugate partitions correspond to anti-symmetric abaci.

$\textcircled{-8}$ $\textcircled{-7}$ $\textcircled{-6}$ -5 $\textcircled{-4}$ -3 -2 $\textcircled{-1}$ $\textcircled{0}$ | 1 2 $\textcircled{3}$ $\textcircled{4}$ 5 $\textcircled{6}$ 7 8 9

Core partitions

The **hook length** of a box = # boxes below + # boxes to right + box
 λ is a **t -core** if no boxes have hook length t .

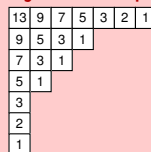
t -core partition



t -flush abacus

○ ○ ○ ○ ○ _ ○ ○ ○ _ _ ○ ○ _ _ ○ _ _ _

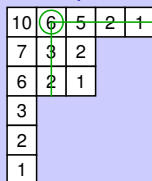
Self-conj. t -core partition



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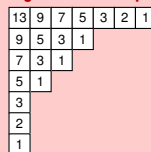
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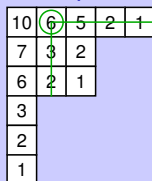
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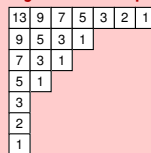
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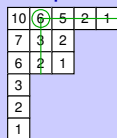
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t -core partition



t -flush abacus (in runners)

⑤ ④ ③ ② ① 0 ① ② ③ 4 5 ⑥ ⑦ 8 9 ⑩ 11 12 13

⑧ ⑦ ⑥ ⑤

⑦ ⑥ ⑤ ④

④ ③ ② ①

③ ② ① ④

0 ① ② ③

① ② ③ ④

4 5 ⑥ ⑦

⑤ ⑥ ⑦ ⑧

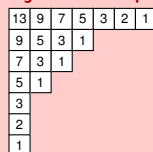
8 9 ⑩ 11

⑨ ⑩ 11 12

Normalized

Balanced

Self-conj. t -core partition



t -flush antisymmetric abacus

⑦ ⑥ ⑤ ④

③ ② ① ④

① ② ③ ④

⑤ ⑥ ⑦ ⑧

9 ⑩ ⑪ 12

Antisymmetry about $t/t + 1$.

Simultaneity

Of interest: Partitions that are **both** s -core **and** t -core. $(s, t) = 1$

- Abaci that are both s -flush and t -flush.

(s, t) -core partitions

9	6	5	3	2	1
5	2	1			
2					
1					

Self-conj. (s, t) -core partitions

9	6	4	2	1
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4	1			
2				
1				

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(Anderson, 2002):

$$\# (s, t)\text{-core partitions} = \frac{1}{s+t} \binom{s+t}{s}$$

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(Ford, Mai, Sze, 2009):

$$\# \text{ self-conj. } (s, t)\text{-core partitions} = \binom{s'+t'}{s'}$$

$$\text{where } s' = \left\lfloor \frac{s}{2} \right\rfloor \text{ and } t' = \left\lfloor \frac{t}{2} \right\rfloor$$

Core partitions in the literature

Representation Theory: (origin)

t -cores label t -blocks of irreducible modular representations for S_n .

Nakayama conj. Brauer-Robinson '47

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$c_t(n) = \#$ of t -core partitions of n .

$$\sum_{n \geq 0} c_t(n) q^n = \prod_{n \geq 1} \frac{(1 - q^{nt})^t}{1 - q^n}$$

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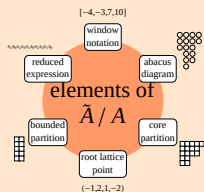
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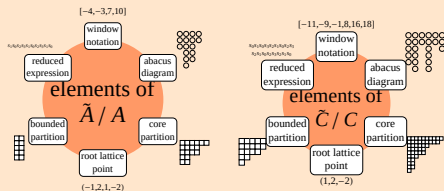
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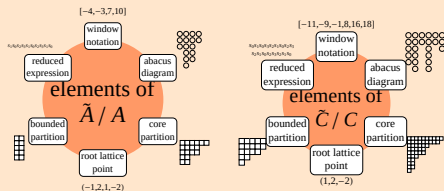
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One interpretation: **Alcove geometry**

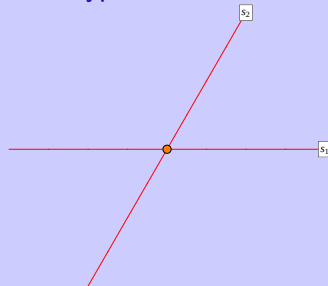
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Type A_t : generators $\{s_1, \dots, s_t\}$

Group of permutations of $\{1, \dots, t+1\}$.

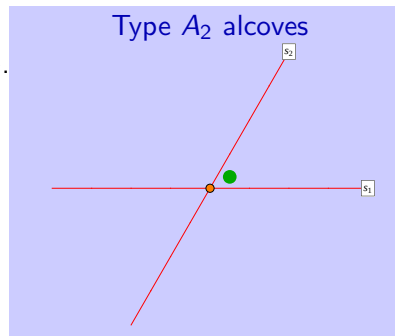
Symmetries of regular simplex, dim. t .

Type A_2 alcoves



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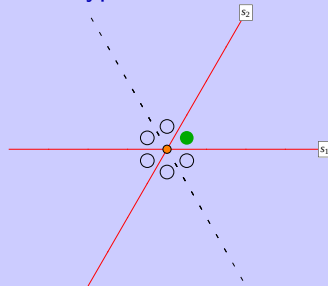
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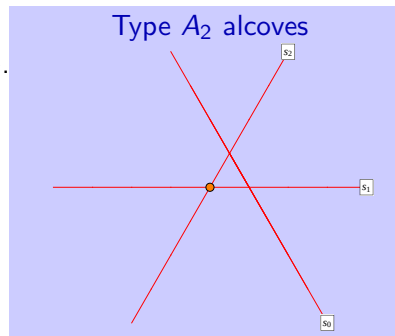
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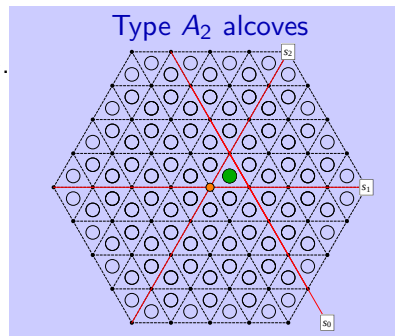
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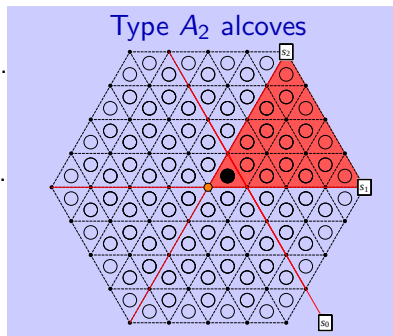
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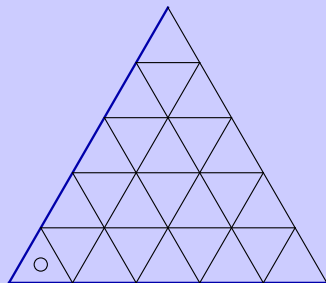
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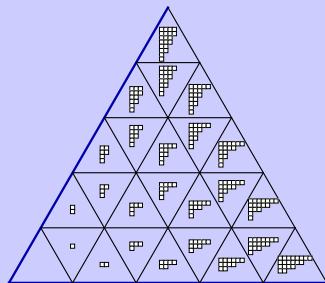
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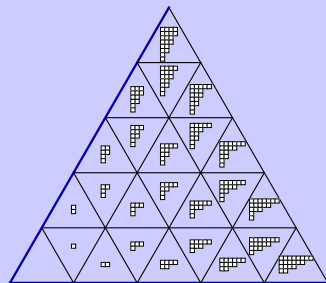
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Overlay the m -Shi arrangement.

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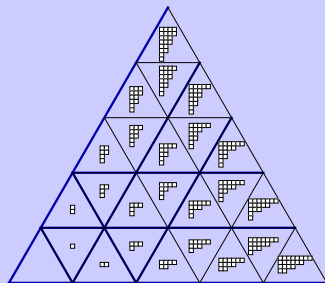
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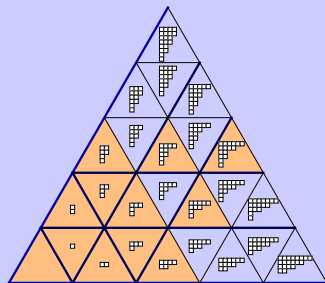
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Which are representative alcoves?

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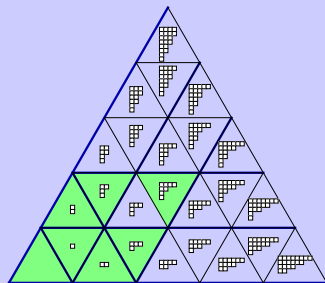
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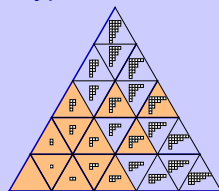
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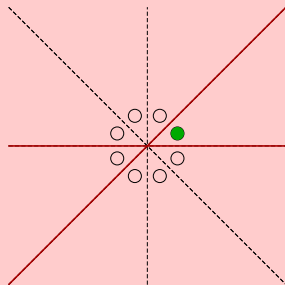
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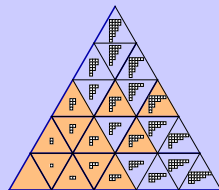
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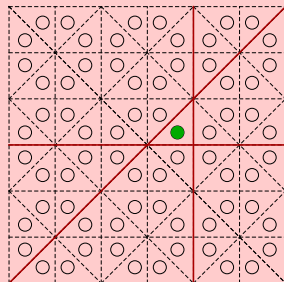
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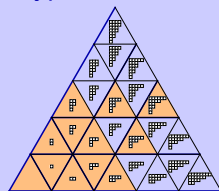
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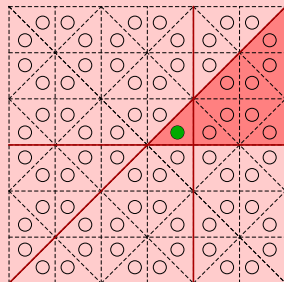
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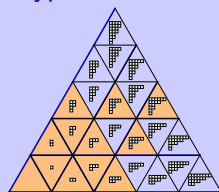
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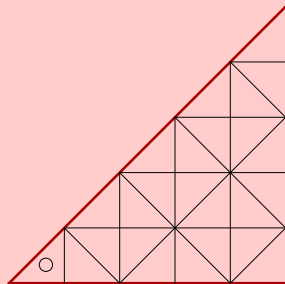
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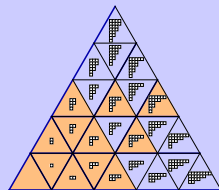
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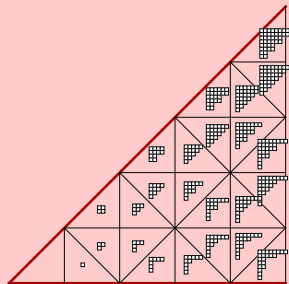
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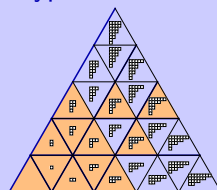
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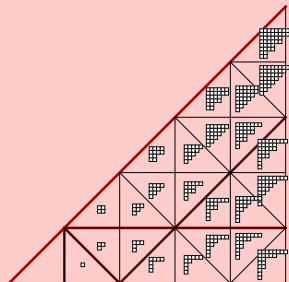
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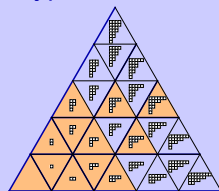
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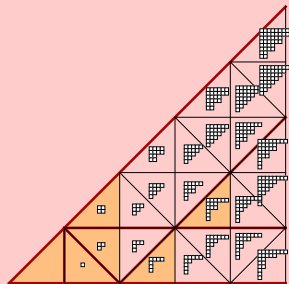
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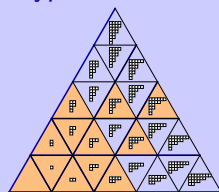
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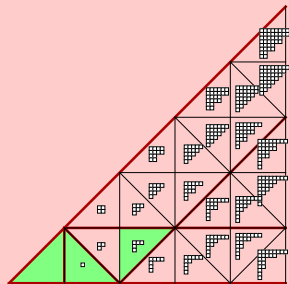
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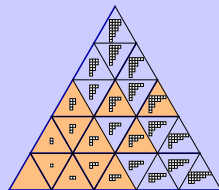
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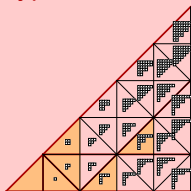
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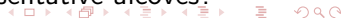
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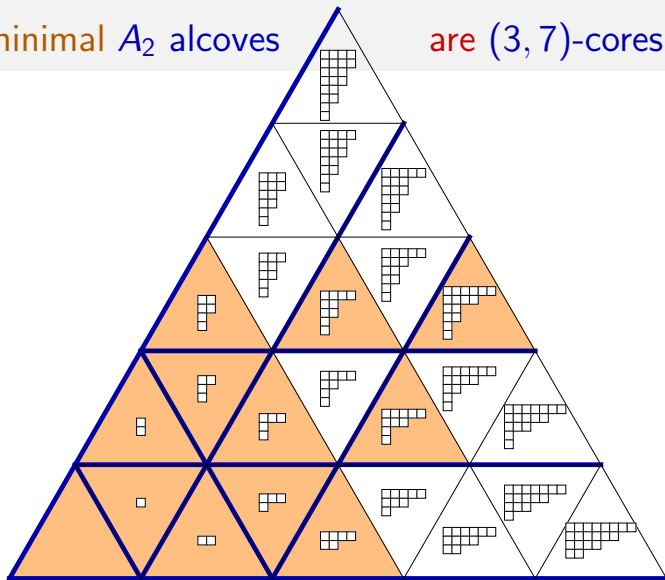
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★ Representative alcoves correspond to simultaneous cores. ★

The 2-minimal A_2 alcoves are $(3, 7)$ -cores



Abaci to the rescue!

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Specializing the results of [Anderson](#) and [Ford, Mai, and Sze](#),

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Question: Is there a simple statistic on simultaneous core partitions that gives us a q -analog of the Catalan numbers?

$$\sum_{\substack{\lambda \text{ is a} \\ (t, t+1)\text{-core}}} q^{\text{stat}(\lambda)} = \frac{1}{[t+1]_q} \begin{bmatrix} 2t \\ t \end{bmatrix}_q$$

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Answer: Yes. We will create an analog of the **major statistic**.

The major statistic

Given a permutation π of $\{1, \dots, n\}$ written in one-line notation as $\pi = \pi_1\pi_2\cdots\pi_n$, the **major statistic** $\text{maj}(\pi)$ is defined as the sum of the positions of the descents of π , in other words,

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Named in honor of Major Percy MacMahon who showed it has the same distribution as the statistic of the number of inversions:

$$\sum_{\pi \in S_n} q^{\text{maj}(\pi)} = \sum_{\pi \in S_n} q^{\text{inv}(\pi)}$$

A major statistic for simultaneous cores

Let λ be a $(t, t+1)$ -core.
Define $b = (b_0, \dots, b_{t-1})$
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Let λ be a **s-c.** $(2t, 2t+1)$ -core.
 Define $b = (b_0, \dots, b_t)$
 where $b_0 = 0$ and $b_i =$
 $(\# \text{ diag. arms } \equiv i \pmod{2t}) -$
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$$\sum_{\substack{\lambda \text{ is a} \\ (t, t+1)\text{-core}}} q^{\text{maj}(\lambda)} = \frac{1}{[t+1]_q} \begin{bmatrix} 2t \\ t \end{bmatrix}_q$$

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Define $b = (b_0, \dots, b_t)$
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$$\text{maj}(\lambda) = 2 \sum_{i: b_{i-1} \geq b_i} (2i - b_i - 1).$$

A major statistic for simultaneous cores

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Note: maj defined as a sum
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A major statistic for abacus diagrams

Let λ be a $(t, t+1)$ -core.

Read off the levels of the defining beads of the (normalized) abacus to give $b = (b_0, \dots, b_{t-1})$.

Define

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Then

$$\sum_{\substack{\lambda \text{ is a} \\ (t, t+1)\text{-core}}} q^{\text{maj}(\lambda)} = \frac{1}{[t+1]_q} \begin{bmatrix} 2t \\ t \end{bmatrix}_q$$

Let λ be a s-c. $(2t, 2t+1)$ -core.

Read off the levels of the defining beads of the corresponding abacus to give $b = (b_0, \dots, b_t)$.

Define

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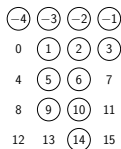
Then

$$\sum_{\substack{\lambda \text{ is a self-conj.} \\ (2t, 2t+1)\text{-core}}} q^{\text{maj}(\lambda)} = \begin{bmatrix} 2t \\ t \end{bmatrix}_{q^2}$$

Proof sketch

- Use Anderson's lattice path bijection:

$$(s, t)\text{-flush abaci} \longleftrightarrow L : (0, 0) \rightarrow (s, t) \text{ above } y = \frac{t}{s}x.$$

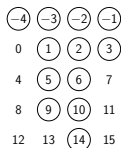


35	31	27	23	19	15	11	7	3	-1	-5	-9	-13
22	18	14	10	6	2	-2	-6	-10	-14	-18	-22	-26
9	5	1	-3	-7	-11	-15	-19	-23	-27	-31	-35	-39
-4	-8	-12	-16	-20	-24	-28	-32	-36	-40	-44	-48	-52

Proof sketch

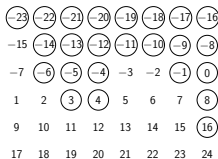
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- Create a similar lattice path bijection: (improves Ford-Mai-Size)
antisymm. (s, t) -flush abaci $\longleftrightarrow L : (0, 0) \rightarrow ([\frac{s}{2}], [\frac{t}{2}])$.

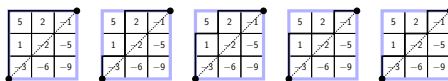


94	86	78	70	62	54	46	38	30	22	14	6	-2
81	73	65	57	49	41	33	25	17	9	1	-7	-15
68	60	52	44	36	28	20	12	4	-4	-12	-20	-28
55	47	39	31	23	15	7	-1	-9	-17	-25	-33	-41
42	34	26	18	10	2	-6	-14	-22	-30	-38	-46	-54
29	21	13	5	-3	-11	-19	-27	-35	-43	-51	-59	-67
16	8	0	-8	-16	-24	-32	-40	-48	-56	-64	-72	-80
3	-5	-13	-21	-29	-37	-45	-53	-61	-69	-77	-85	-93

Proof sketch

- $(t, t + 1)$ -flush abaci $\longleftrightarrow L : (0, 0) \rightarrow (t, t)$ above $y = x$.

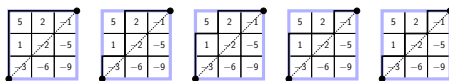
Dyck paths!



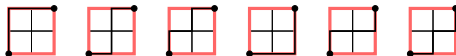
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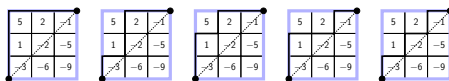
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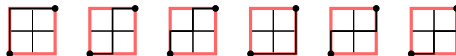
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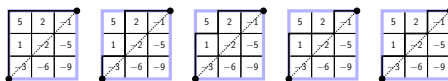
- Use the major index on lattice paths that is known to give the desired q -analog:

$$\text{maj}(L) = \sum_{i:(L_i, L_{i+1})=(E, N)} i$$

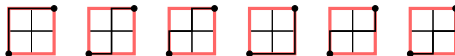
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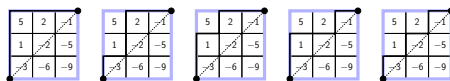
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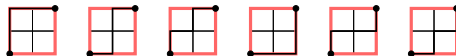
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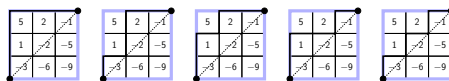
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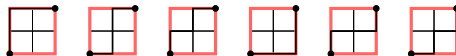
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- Translate this major index to language of abaci and cores.

Talk Recap

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 - ▶ Core partitions and abacus diagrams
 - ▶ Simultaneity

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- ▶ Remarkable
 - ▶ Type-independent setup.
 - ▶ Abaci are the right tool.

What's next?

1. Core survey

- ▶ Compile combinatorial interpretations into illustrated dictionary.
- ▶ Reconcile many appearances of cores into historical survey.
- ▶ Gathering sources stage — What do you know?

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1. Core survey
2. Open question: Catalan q -analogs

► **Question.** Is there a core statistic for a q -analog of $\frac{1}{a+b} \binom{a+b}{a}$?

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- ▶ **Progress:** Drew has a candidate statistic.

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4. Open question: Cyclic sieving phenomenon
 - ▶ Note: $\frac{1}{[a+b]_q} \begin{bmatrix} a+b \\ a \end{bmatrix}_q \Big|_{q=-1} = \binom{\lfloor \frac{a}{2} \rfloor + \lfloor \frac{b}{2} \rfloor}{\lfloor \frac{a}{2} \rfloor}$.

Thank you!

Slides available: people.qc.cuny.edu/chanusa > **Talks**

Interact: people.qc.cuny.edu/chanusa > **Animations**



Drew Armstrong, Christopher R. H. Hanusa, Brant C. Jones.
Results and conjectures on simultaneous core partitions.
Submitted, 2013. arXiv:1308.0572.



Christopher R. H. Hanusa and Brant C. Jones.
Abacus models for parabolic quotients of affine Coxeter groups
Journal of Algebra. Vol. 361, 134–162. (2012) arXiv:1105.5333



Gordon James and Adalbert Kerber.
The representation theory of the symmetric group.
Addison-Wesley, 1981.